

Madison Action Layer

AI-routed handoff from content to CRM: every ad variant is checked, decided and turned into real work in about 7 seconds. The pitch first, then the proof: 11 annotated screenshots from real runs on 2026-10-09.

★ The pitch

- 1 The workflow: four agents in one n8n canvas
- 3 Never trust the AI blindly: the validator
- 5 The validator rejects the failed AI answer
- 7 Output: an alert a busy marketer can act on
- 9 Visibility: the shared decision log
- 11 Reliability: duplicates are not re-processed

- 2 Intelligence: Claude's routing decision
- 4 Failing safely: Claude unreachable, work still routed
- 6 ...and the team is told the AI was skipped
- 8 Action: tasks land in the CRM
- 10 Scale: 400 requests, 100 at a time
- What's next

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Stack n8n 2.12.3 (self-hosted) + Claude Haiku 4.5

Integrations HubSpot · Slack · Google Sheets · Gmail

THE PITCH

AI already writes the ads. **Madison Action Layer** makes sure they ship.

Every new ad variant is checked, judged by Claude and turned into a HubSpot task and a Slack alert for the right team in about 7 seconds, for less than half a cent. People only step in when their judgement is actually needed.

THE PROBLEM

Madison's Content Agent writes and quality-checks ad variants, and then they sit in a CSV. Someone still has to read each one, decide, file a CRM task and tell the team. Marketers lose about **10 hours a week (26% of their time)** to handoffs like this (DoubleVerify 2025, 1,970 marketers).

THE SOLUTION

An action layer between AI content and the people who ship it. Claude scores each ad against its channel's real limits, flags risks with the exact words, decides **approve, review or block**, picks the owner, and writes the fix, the task and the alert.

WHY YOU CAN TRUST IT

Claude must answer in a strict schema, and every answer is validated before anything is sent. If the AI fails, a rule-based fallback routes the ad to a human. Tested live with Claude switched off: **no crash, no lost ad**.

NO AI

① Gatekeeper

Blocks broken, low-quality and duplicate ads in 0.2 s, at no AI cost.

CLAUDE HAIKU 4.5

② Routing Agent

Channel fit, risk flags with evidence, decision, owner and a rewrite.

INTEGRATIONS

③ Execution Agent

HubSpot task + Slack alert, retried and logged, every time.

CLAUDE HAIKU 4.5

④ Insights Agent

Daily report: patterns, recommendations and a watchlist.

834

real requests in load tests

0

failed requests

\$0.0044

AI cost per ad

6.4 s

median webhook → Slack + HubSpot

≈330/min

on a single 1-vCPU server

One real batch of 50 ads: **\$0.16 of AI replaced ≈3.5 hours of manual handoff (≈\$121)**. At 1,000 ads a day that is ≈\$95–135 a month in AI costs, on the server already running, with no new software seats.

BEYOND THE BRIEF: PROBLEMS WE FOUND AND SOLVED

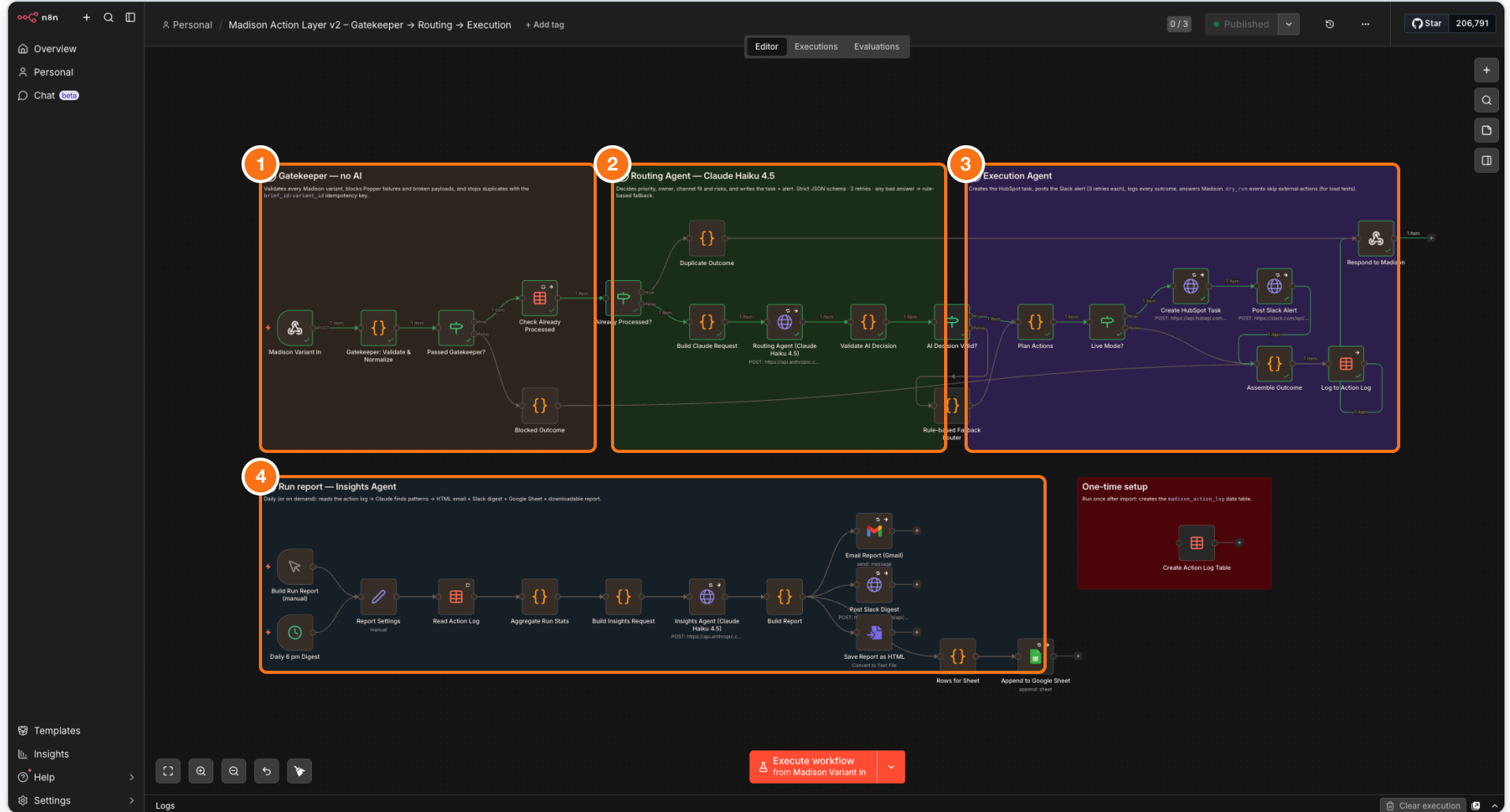
An AI that blocked too much. The first prompt blocked 85% of ads. Calibrating severity gave a realistic mix (16 approved · 18 review · 1 blocked) that was faster (p50 8.9 s → 6.4 s) and cheaper.

The real bottleneck. Under load the limit was n8n's intake on 1 vCPU (requests queued ~7 s), not Claude: 0 rate-limit errors at 100 concurrent requests.

A hidden duplicate race. Sequential repeats are caught in 0.19 s at \$0, but simultaneous duplicates slip through. Found under load, documented with evidence, fix designed.

The workflow: four agents in one n8n canvas

One webhook in, real actions out. Each event only runs the branch it needs — blocked, duplicate and fallback branches stay grey.



- ① Gatekeeper (no AI): schema, date and Popper checks + idempotency lookup.
- ③ Execution Agent: HubSpot task → Slack alert → log → answer Madison.

- ② Routing Agent: Claude Haiku 4.5 with a strict JSON schema, then a validator.
- ④ Insights Agent: daily/on-demand batch report → email, Slack, Sheet, HTML.

Intelligence: Claude's routing decision

The Routing Agent receives the variant, the campaign context and market context, and must answer in a fixed schema.

The screenshot displays the Routing Agent interface (Claude Haiku 4.5) with four numbered callouts highlighting key components:

- System Prompt:** Located in the 'INPUT' section under 'request_body' > 'system'. It defines the agent's role as a routing agent in Madison's Action Layer, responsible for deciding what happens to each variant before it reaches the team's CRM and chat. It includes rules for channel fit (score 1-5) and character limits for Google Ads and Facebook.
- JSON Schema:** Located in the 'INPUT' section under 'output_config' > 'format' > 'schema'. It defines the required fields for the output, including decision, priority, owner_team, slack_channel, channel_fit, risk_flags, rationale, suggested_fix, hubspot_task, slack_message, and confidence.
- Output:** Located in the 'OUTPUT' section. It shows the JSON response from the agent, including the decision (needs_review), priority (normal), owner_team (Social & Influencer), slack_channel (campaign-review), channel_fit (score: 4), note (Strong hook in first 40 characters), risk_flags (false_urgency, low), evidence (Limited stock - shop now!), type (message), role (assistant), and content (a detailed rewrite of the variant copy).
- Usage:** Located in the 'OUTPUT' section under 'usage'. It shows the token usage for the request: input_tokens: 2199, cache_creation_input_tokens: 0, cache_read_input_tokens: 0, ephemeral_5m_input_tokens: 0, ephemeral_1h_input_tokens: 0, and output_tokens: 474.

- System prompt: channel limits, severity calibration, decision + priority rules.
- JSON schema with enums — the answer is machine-checkable, not free text.
- Decision: needs_review, owner, channel fit 4/5, two risk flags with evidence, a rewrite.
- Real usage: 2,199 input + 474 output tokens.

Never trust the AI blindly: the validator

Every answer is checked before anything is sent. Refusals, truncation, bad JSON or invalid values go to a rule-based fallback.

The screenshot displays the 'Validate AI Decision' interface, which is divided into several sections: INPUT, PARAMETERS, SETTINGS, and OUTPUT. The interface is annotated with four numbered circles (1-4) highlighting specific features.

INPUT Section: Shows a routing agent (Claude Haiku 4.5) with a message ID and content. The content is a JSON object with a 'text' field containing a decision and a 'note'.

PARAMETERS Section: Shows the 'Mode' set to 'Run Once for All Items' and the 'Language' set to 'JavaScript'. A note at the bottom says 'Type \$ for a list or special vars/methods. Debug by using console.log() statements and viewing their output in the browser console.'

SETTINGS Section: Shows the 'JavaScript' code for the validator, which checks for API errors, refusals, truncation, and malformed JSON. It also checks for out-of-range values and uses a rule-based fallback.

OUTPUT Section: Shows the result of the validation, including the 'ai_ok' status, the 'decision' object, and the 'risk_flags' array. The 'ai_ok' status is 'true', indicating the answer passed every check. The 'decision' object contains the 'text' field from the input. The 'risk_flags' array contains two items, each with a 'type', 'severity', and 'evidence'.

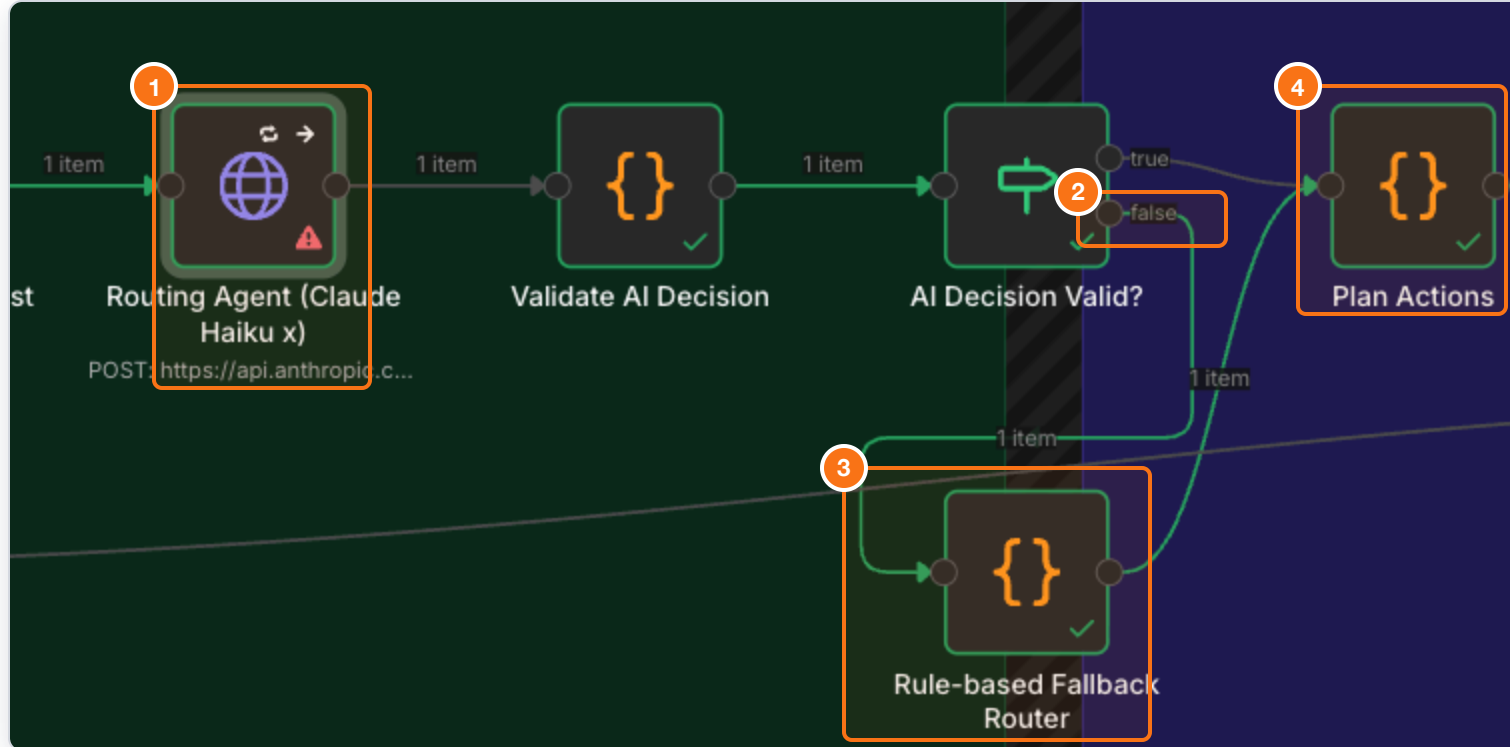
Annotations:

- 1:** Points to the 'ai_ok' status in the output, which is 'true'.
- 2:** Points to the 'content' field in the input, which contains the JSON object with the 'text' field.
- 3:** Points to the 'risk_flags' array in the output, which contains two items with 'type', 'severity', and 'evidence'.
- 4:** Points to the 'cost_usd' field in the output, which is '0.004569'.

- 1** ai_ok: true — the answer passed every check.
- 2** Checks: API error, refusal, max_tokens, JSON parse, enum values, required text.
- 3** Risk flags carry the exact evidence and a severity.
- 4** Cost tracked per event: \$0.0046, latency 6.7 s.

Failing safely: Claude unreachable, work still routed

Live test fallback-1: model ID set to an invalid value. The path bends through the fallback router instead of stopping.



- 1** Routing Agent errors (⚠) after 3 retries — but continues.
- 3** Rule-based Fallback Router: regex risks + team by channel.

- 2** Validator answers false → branch to fallback.
- 4** Plan Actions continues: task + Slack alert still sent.

The validator rejects the failed AI answer

Validate AI Decision receives Claude's API error instead of a decision, marks it invalid and hands the event to the fallback router.

INPUT

- Routing Agent (Claude Haiku x) 1 item
 - error
 - message: The resource you are requesting could not be found
 - timestamp: 1791550163538
 - name: NodeApiError
 - description: model: claude-haiku-x
 - context
- Build Claude Request 1 item
- Already Processed?
- Check Already Processed Preview
- Passed Gatekeeper? 1 item
- Gatekeeper: Validate & Normalize 1 item
- Madison Variant In 1 item

Parameters

Mode: Run Once for All Items

Language: JavaScript

JavaScript

```
// ② ROUTING AGENT - answer checker. Never trusts the AI blindly:
// API errors (rate limit / timeout / 5xx after 3 retries),
// out-of-range values all mark the decision invalid, so the
// rule-based fallback takes over.
const res = $input.first().json;
const started = $('Build Claude Request').first().json.ai_started_at_ms;
const PRICE_IN = 1.0 / 1e6, PRICE_OUT = 5.0 / 1e6; // Claude Haiku 4.5: $1 / $5 per million tokens

const ENUMS = {
  decision: ['auto_approve', 'needs_review', 'block'],
  priority: ['urgent', 'high', 'normal', 'low'],
  owner_team: ['Paid Media', 'Lifecycle & Email', 'Social & Influencer', 'Content & SEO', 'Brand & Legal Review'],
  slack_channel: ['campaign-urgent', 'campaign-review', 'campaign-digest'],
};

let problem = null;
let decision = null;
if (res.error) {
```

OUTPUT

- ai_ok: false
- ai_problem: api_error: The resource you are requesting could not be found
- decision: [null]
- ai_model: claude-haiku-4-5
- ai_latency_ms: 10950
- input_tokens: 0
- output_tokens: 0
- cost_usd: 0

Type \$ for a list of [special vars/methods](#). Debug by using `console.log()` statements and viewing their output in the browser console.

- 1 Claude's error arrives as data (model claude-haiku-x not found) after 3 retries.
- 3 Allowed values for decision, priority, owner team and Slack channel are enforced.

- 2 ai_ok: false + ai_problem: api_error, so the answer is rejected.
- 4 No tokens used: \$0 AI cost for this event.

...and the team is told the AI was skipped

The fallback alert in #campaign-review — a human checks it; nothing is lost.

The screenshot shows a Slack interface for the 'Madison Project' workspace. The left sidebar contains navigation options like Home, DMs, Activity, Files, Channels, Agents & Tools, and Admin. The main area shows the '# campaign-review' channel. A message from 'Madison Bot' is displayed, containing a HubSpot task and a fallback alert. Three orange circles with numbers 1, 2, and 3 highlight specific parts of the message:

- 1** Points to the message header: **Madison Bot** APP · 08:49 · **NORMAL** · **Needs review**.
- 2** Points to the 'Channel fit' section: **Channel fit** 3/5 — Not assessed (AI unavailable).
- 3** Points to the 'Suggested fix' section: **Suggested fix** Spot task #405971552959 · **Routed by fallback rules** · confidence 0 · evt_fallback-1_0001.

The message content includes a HubSpot task for 'Discover Harem Pants!' and a fallback alert indicating that the AI router is unavailable and a manual check is required.

1 ⚠️ AI router unavailable — with the real API error.

2 Channel fit: not assessed (AI unavailable).

3 HubSpot task created · 'Routed by fallback rules' · confidence 0.

Output: an alert a busy marketer can act on

Slack Block Kit message in #campaign-review — decision, owner, channel fit, evidence and the fix, linked to the HubSpot task.

The screenshot shows a Slack interface with a sidebar on the left containing navigation options like Home, DMs, Activity, Files, Agents & tools, and Admin. The main content area displays a message in the #campaign-review channel. The message is a HubSpot task review for 'Alpha Innovations email variant'. It includes a decision 'NORMAL' with a 'Needs review' status, an owner 'Lifestyle & Email', and a channel fit of '2/5'. The message also contains risk flags, a suggested fix, and a HubSpot task ID. Four numbered callouts are present: 1 points to the risk flags, 2 points to the suggested fix, 3 points to the channel fit, and 4 points to the HubSpot task ID.

Message content:

viewing CTA, and keeps the edgy tone.

HubSpot task #405966806771 · Routed by Claude Haiku 4.5 · confidence 0.78 · evt_v2-50_0039

NORMAL · Needs review

Alpha Innovations email variant needs subject line trim (87→60 chars max). Copy is strong; fix headline length and we're good to ship.

Campaign	Owner
Alpha Innovations · Email on Email	Lifestyle & Email

Headline
Discover elegance redefined with our timeless Boatneck Blouse - the perfect blend of a classic wide neckline & feminine charm, crafted for your ultima

Channel fit
2/5 — Headline exceeds 60-character email subject line limit (87 chars). Current phrasing is too long and will truncate, losin

Risk flags

- unverifiable claim (low): "ultimate style statement" and "elegance redefined" are standard marketing adjectives, not false claims; acceptable for fashion copy.
- weak or missing cta (low): "Shop now!" is generic and doesn't tie to the product benefit (e.g., "Get your Boatneck Blouse today" or "Discover timeless elegance").

Suggested fix
Headline: 'Timeless Boatneck Blouse – Elegance Redefined' (49 chars). CTA: 'Shop the Boatneck Blouse today' to tie action to product.

HubSpot task #405966335726 · Routed by Claude Haiku 4.5 · confidence 0.78 · evt_v2-50_0041

06:57 **NORMAL** · Needs review

⚠️ Chunky Knit Sweater (AD-0018-v2-50-0042) needs review: headline 53/30 chars, 'unbeatable' claim flagged, casual tone misaligned with Google Ads. Route to Paid Media for edit.

Campaign	Owner
Innovate Industries · Search on Google Ads	Paid Media

Headline
Stay warm this season with our Chunky Knit Sweater!

Channel fit
2/5 — Headline is 53 characters (limit 30); body is 89 characters (acceptable); copy uses casual tone with emoji-like language

Risk flags

- unverifiable claim (medium): "Unbeatable comfort" is a comparative superlative without substantiation; typically flagged as unproven marketing claim in Google Ads policy.
- channel mismatch (medium): "Google Ads search copy should be direct and benefit-focused; 'snuggle up' is colloquial and the CTA tone doesn't match Google Ads conventions for this demographic (Men 18-24 - Foodies)."

Suggested fix
Headline: 'Chunky Knit Sweater – Cozy & Warm' (34 chars, trim to 30). Body: 'Thick, textured knit perfect for layering. Premium comfort and timeless style!' CTA: 'Shop now.'

HubSpot task #405887122139 · Routed by Claude Haiku 4.5 · confidence 0.72 · evt_v2-50_0042

Message #campaign-review

1 Risk flags quote the copy ('Unbeatable comfort').

3 Channel fit 2/5 with the reason (53 chars vs 30).

2 Suggested fix: a headline that fits the 30-char Google Ads limit.

4 HubSpot task ID + 'Routed by Claude Haiku 4.5' + confidence.

Action: tasks land in the CRM

Each routed variant becomes a HubSpot task with an imperative title, priority and due date (urgent = today).

The screenshot shows the HubSpot Tasks interface. The left sidebar contains navigation links for Home, Contacts, Companies, Deals, Tasks, Sales, Activity Feed, Prospecting Agent, Sales Workspace, Platform, and Dashboards. The main area displays a list of tasks with columns for Task Title, Task stage, Task → Contacts, Task → Companies, Last contacted, Last engagement, Task Type, and Due date (EDT). Three orange boxes highlight specific columns: 1. Task Title, 2. Task stage, and 3. Due date (EDT).

Task Title	Task stage	Task → Contacts	Task → Companies	Last contacted	Last engagement	Task Type	Due date (EDT)
Block & investigate ...	Not Started	--	--	--	--	To-do	Today at 6:22 AM EDT
Investigate & rerout...	Not Started	--	--	--	--	To-do	Today at 6:31 AM EDT
Block & investigate ...	Not Started	--	--	--	--	To-do	Today at 6:32 AM EDT
Resolve brief mism...	Not Started	--	--	--	--	To-do	Today at 6:32 AM EDT
Block: Fix campaig...	Not Started	--	--	--	--	To-do	Today at 6:33 AM EDT
Block & clarify: Gla...	Not Started	--	--	--	--	To-do	Today at 6:33 AM EDT
Resolve brand mis...	Not Started	--	--	--	--	To-do	Today at 6:33 AM EDT
Clarify campaign ro...	Not Started	--	--	--	--	To-do	Today at 6:33 AM EDT
Block & reassign: H...	Not Started	--	--	--	--	To-do	Today at 6:33 AM EDT
Block & clarify: Gla...	Not Started	--	--	--	--	To-do	Today at 6:33 AM EDT
URGENT: Block AD...	Not Started	--	--	--	--	To-do	Today at 6:33 AM EDT
Block: Resolve Tass...	Not Started	--	--	--	--	To-do	Today at 6:34 AM EDT
URGENT: Block har...	Not Started	--	--	--	--	To-do	Today at 6:34 AM EDT
BLOCK: Resolve ch...	Not Started	--	--	--	--	To-do	Today at 6:34 AM EDT
Reject & reroute Yo...	Not Started	--	--	--	--	To-do	Today at 6:34 AM EDT
Publish email varia...	Not Started	--	--	--	--	To-do	Today at 6:52 AM EDT
Publish Instagram v...	Not Started	--	--	--	--	To-do	Today at 6:52 AM EDT
Publish Google Ads ...	Not Started	--	--	--	--	To-do	Today at 6:52 AM EDT
Publish Google Ads ...	Not Started	--	--	--	--	To-do	Today at 6:57 AM EDT
Publish email varia...	Not Started	--	--	--	--	To-do	Today at 6:57 AM EDT
Publish Instagram v...	Not Started	--	--	--	--	To-do	Today at 6:57 AM EDT
Publish Google Ads ...	Not Started	--	--	--	--	To-do	Today at 6:57 AM EDT
Publish embroidere...	Not Started	--	--	--	--	To-do	Today at 6:57 AM EDT
Publish Duster Coat...	Not Started	--	--	--	--	To-do	Today at 6:57 AM EDT
Publish Bolero Jack...	Not Started	--	--	--	--	To-do	Today at 6:57 AM EDT
Publish choker nec...	Not Started	--	--	--	--	To-do	Today at 6:57 AM EDT
Block: Resolve brief...	Not Started	--	--	--	--	To-do	Tomorrow at 6:32 AM EDT
Block: Fix campaig...	Not Started	--	--	--	--	To-do	Tomorrow at 6:32 AM EDT
Resolve brief/chan...	Not Started	--	--	--	--	To-do	Tomorrow at 6:32 AM EDT
Block & realign: Tas...	Not Started	--	--	--	--	To-do	Tomorrow at 6:32 AM EDT
Block AD-0029: Red...	Not Started	--	--	--	--	To-do	Tomorrow at 6:33 AM EDT

- Imperative task titles written by Claude.
- All created via the HubSpot API — status Not Started.
- Due date set from priority (today / tomorrow / 3 days).

Visibility: the shared decision log

Google Sheet appended in one batch per report — readable without opening n8n.

logged_at	run_id	event_id	headline	decision	priority	owner_team	status	ai	skipped	risk_count	why	spot_task_id	slack_poste	latency_s	cost_usd
2026-10-09 10:5 v2-50		evt_v2-50_0004		block	low	Gatekeeper	blocked	skipped		1	Blocked by Gatekeeper: missing		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0007	Stay warm, loo	block	low	Gatekeeper	blocked	skipped		1	Blocked by Gatekeeper: invalid_c		no	0.2	0
2026-10-09 10:5 v2-50		evt_v2-50_0006	Upgrade your v	auto_approve	high	Lifecycle & Email	completed	ai		0	Strong email var	405966640871	yes	6.5	0.003861
2026-10-09 10:5 v2-50		evt_v2-50_0008	Discover elega	auto_approve	urgent	Paid Media	completed	ai		1	Strong performa	405979909865	yes	6.6	0.004231
2026-10-09 10:5 v2-50		evt_v2-50_0014		block	low	Gatekeeper	blocked	skipped		3	Blocked by Gatekeeper: missing		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0002	Introducing the	auto_approve	high	Social & Influenc	completed	ai		0	No HIGH or MET	405986992882	yes	6.9	0.004005
2026-10-09 10:5 v2-50		evt_v2-50_0009	Introducing Ciga	auto_approve	normal	Social & Influenc	completed	ai		0	The variant is we	405963677375	yes	7.5	0.004044
2026-10-09 10:5 v2-50		evt_v2-50_0011	Discover elega	auto_approve	normal	Social & Influenc	completed	ai		1	Copy is well-su	405966806764	yes	6.6	0.004097
2026-10-09 10:5 v2-50		evt_v2-50_0017	Discover our sil	block	low	Gatekeeper	blocked	skipped		1	Blocked by Gatekeeper: invalid_c		no	0.1	0
2026-10-09 10:5 v2-50		evt_v2-50_0012	Step into style	needs_review	normal	Social & Influenc	completed	ai		2	HIGH-severity pl	405987930861	yes	6.7	0.00447
2026-10-09 10:5 v2-50		evt_v2-50_0001	Discover Harer	needs_review	normal	Social & Influenc	completed	ai		2	Variant is publi	405966581439	yes	8.3	0.004676
2026-10-09 10:5 v2-50		evt_v2-50_0021	Introducing our	block	low	Gatekeeper	blocked	skipped		2	Blocked by Gatekeeper: missing		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0005	Elevate your st	needs_review	high	Social & Influenc	completed	ai		1	Strong performa	405859666651	yes	8.7	0.004692
2026-10-09 10:5 v2-50		evt_v2-50_0003	Discover the per	needs_review	normal	Social & Influenc	completed	ai		1	The variant has	405859666652	yes	8.8	0.004668
2026-10-09 10:5 v2-50		evt_v2-50_0024		block	low	Gatekeeper	blocked	skipped		1	Blocked by Gatekeeper: missing		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0010	Introducing Gloc	needs_review	high	Social & Influenc	completed	ai		2	Strong performa	405924769477	yes	9.2	0.004965
2026-10-09 10:5 v2-50		evt_v2-50_0018	Introducing the	auto_approve	normal	Social & Influenc	completed	ai		0	No HIGH or MET	405987701467	yes	4.8	0.004019
2026-10-09 10:5 v2-50		evt_v2-50_0022	Upgrade your v	auto_approve	normal	Lifecycle & Email	completed	ai		0	Clean email vari	405966677712	yes	5.7	0.003959
2026-10-09 10:5 v2-50		evt_v2-50_0027	Introducing the	block	low	Gatekeeper	blocked	skipped		1	Blocked by Gatekeeper: invalid_c		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0016	Discover the co	needs_review	normal	Social & Influenc	completed	ai		1	The variant has	405924769483	yes	8.1	0.004763
2026-10-09 10:5 v2-50		evt_v2-50_0015	Discover unpar	needs_review	normal	Lifecycle & Email	completed	ai		2	The headline ex	405966460665	yes	8.2	0.00485
2026-10-09 10:5 v2-50		evt_v2-50_0029	Discover the per	block	low	Gatekeeper	blocked	skipped		2	Blocked by Gatekeeper: missing		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0013	Discover unpar	needs_review	normal	Social & Influenc	completed	ai		2	Channel fit is ac	405966334661	yes	8.8	0.004894
2026-10-09 10:5 v2-50		evt_v2-50_0023	Discover our too	auto_approve	high	Paid Media	completed	ai		0	Strong conversi	405862844121	yes	6.7	0.004233
2026-10-09 10:5 v2-50		evt_v2-50_0025	Introducing the	needs_review	normal	Social & Influenc	completed	ai		1	Variant has no H	405987901166	yes	7.2	0.004779
2026-10-09 10:5 v2-50		evt_v2-50_0020	Discover high-v	needs_review	normal	Social & Influenc	completed	ai		2	Copy is well-writ	405960284879	yes	8.1	0.00485
2026-10-09 10:5 v2-50		evt_v2-50_0034		block	low	Gatekeeper	blocked	skipped		1	Blocked by Gatekeeper: missing		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0019	Discover timeles	needs_review	normal	Paid Media	completed	ai		2	Headline violat	405873973953	yes	8.1	0.005059
2026-10-09 10:5 v2-50		evt_v2-50_0036	Introducing our	block	low	Gatekeeper	blocked	skipped		2	Blocked by Gatekeeper: missing		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0037	Step into elega	block	low	Gatekeeper	blocked	skipped		1	Blocked by Gatekeeper: invalid_c		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0038	Reveal your cas	block	low	Gatekeeper	blocked	skipped		2	Blocked by Gatekeeper: missing		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0026	Refine your sty	needs_review	normal	Social & Influenc	completed	ai		2	The variant has	405960284881	yes	7.7	0.004699
2026-10-09 10:5 v2-50		evt_v2-50_0032	Discover stunn	auto_approve	high	Content & SEO	completed	ai		0	Strong performa	405966769894	yes	4.9	0.003983
2026-10-09 10:5 v2-50		evt_v2-50_0026	Refine your sty	needs_review	normal	Social & Influenc	completed	ai		1	The variant is te	405987796725	yes	8	0.004814
2026-10-09 10:5 v2-50		evt_v2-50_0031	Discover chic co	auto_approve	normal	Social & Influenc	completed	ai		0	Variant is well-cr	405987901166	yes	5.9	0.003951
2026-10-09 10:5 v2-50		evt_v2-50_0044		block	low	Gatekeeper	blocked	skipped		1	Blocked by Gatekeeper: missing		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0045	Discover our tim	block	low	Gatekeeper	blocked	skipped		2	Blocked by Gatekeeper: missing		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0033	Discover timeles	needs_review	normal	Social & Influenc	completed	ai		2	HIGH-severity pl	405966461630	yes	6.2	0.004333
2026-10-09 10:5 v2-50		evt_v2-50_0028	Discover sleek s	block	high	Brand & Legal R	completed	ai		3	Variant must be	405966677720	yes	7.2	0.005173
2026-10-09 10:5 v2-50		evt_v2-50_0035	Introducing our	auto_approve	normal	Social & Influenc	completed	ai		1	Variant is well-su	405885554421	yes	5.6	0.004481
2026-10-09 10:5 v2-50		evt_v2-50_0047	Upgrade your v	block	low	Gatekeeper	blocked	skipped		1	Blocked by Gatekeeper: invalid_c		no	0	0
2026-10-09 10:5 v2-50		evt_v2-50_0030	Upgrade your cu	needs_review	normal	Social & Influenc	completed	ai		2	Channel fit is ad	405987932888	yes	7.8	0.004723
2026-10-09 10:5 v2-50		evt_v2-50_0040	Discover unpar	auto_approve	normal	Social & Influenc	completed	ai		0	Variant is well-cr	405901647589	yes	6.3	0.004338
2026-10-09 10:5 v2-50		evt_v2-50_0039	Discover the ult	needs_review	high	Social & Influenc	completed	ai		1	Strong performa	405966806771	yes	6.9	0.004854
2026-10-09 10:5 v2-50		evt_v2-50_0041	Discover elega	needs_review	normal	Lifecycle & Email	completed	ai		2	Email subject lin	405966335726	yes	7.1	0.004579
2026-10-09 10:5 v2-50		evt_v2-50_0043	Introducing our	auto_approve	normal	Social & Influenc	completed	ai		0	No HIGH or MET	405987901173	yes	5.6	0.004106
2026-10-09 10:5 v2-50		evt_v2-50_0042	Stay warm this	needs_review	normal	Paid Media	completed	ai		2	Headline exceed	405987122139	yes	6.5	0.004653
2026-10-09 10:5 v2-50		evt_v2-50_0048	Introducing our	auto_approve	normal	Lifecycle & Email	completed	ai		0	Clean email vari	405987932890	yes	5	0.003945
2026-10-09 10:5 v2-50		evt_v2-50_0049	Discover the ver	auto_approve	high	Content & SEO	completed	ai		0	Strong conversi	405949138657	yes	5	0.003875
2026-10-09 10:5 v2-50		evt_v2-50_0046	Revolutionize yo	auto_approve	high	Content & SEO	completed	ai		0	Strong variant w	405960040660	yes	5.9	0.003955

1 Decision mix: auto_approve / needs_review / block.

2 Routed by Claude (ai) or skipped by the Gatekeeper.

3 HubSpot task ID — traceable to the CRM.

4 Latency and cost per decision.

Scale: 400 requests, 100 at a time

Real HTTP load against the production webhook on a 1 vCPU VPS. Full results: [scale_test_results.md](#).

```
mayank@Mayanks-MacBook-Air Assignment_4 % cat scale_test/results/v2-400.json
{
  "label": "v2-400",
  "requests": 400,
  "concurrency": 100,
  "wall_clock_s": 71.8,
  "throughput_per_min": 334.1,
  "latency_s": {
    "min": 2.544,
    "p50": 16.302,
    "p95": 26.819,
    "max": 31.503,
    "mean": 15.66
  },
  "http_status": {
    "200": 260,
    "422": 140
  },
  "status": {
    "blocked": 140,
    "completed": 260
  },
  "ai_status": {
    "ai": 260,
    "skipped": 140
  },
  "failures": 0,
  "ai_cost_usd": 1.1659,
  "ai_cost_per_routed_event_usd": 0.00448,
  "sample_errors": []
}
```

1 400 requests · 100 concurrent · 71.8 s · 334/min.

3 0 HTTP errors: 260 routed, 140 correctly rejected (422).

2 Latency p50 16 s / p95 27 s — the saturation point (≤ 50 concurrent: ~ 9 s).

4 Total AI cost \$1.17 → \$0.0045 per routed event.

Reliability: duplicates are not re-processed

Re-sending an already-processed event is answered from the action log — no AI call, no task, no alert.

```
mayank@Mayanks-MacBook-Air Assignment_4 % cat scale_test/results/dup-check.json
{
  "label": "dup-check",
  "requests": 1,
  "concurrency": 1,
  "wall_clock_s": 0.2,
  "throughput_per_min": 304.8,
  "latency_s": {
    "min": 0.194,
    "p50": 0.194,
    "p95": 0.194,
    "max": 0.194,
    "mean": 0.19
  },
  "http_status": {
    "200": 1
  },
  "status": {
    "duplicate": 1
  },
  "ai_status": {
    "": 1
  },
  "failures": 0,
  "ai_cost_usd": 0,
  "ai_cost_per_routed_event_usd": 0.0,
  "sample_errors": []
}
```

1 status: duplicate

3 AI cost \$0

2 0.19 s end to end

WHAT'S NEXT

From content that waits to content that moves.

Proven today

- Real outputs in HubSpot, Slack, Gmail and Google Sheets, from one webhook call per ad.
- 834 real requests, up to 100 at once, with 0 failures and 0 lost events.
- Every decision explained: risk flags quote the copy, and every fix is written out.
- Fails safely: retries, an answer validator, a fallback router and crash alerts.
- Cost and latency measured for every single event.

Where it goes next

- **Multi-team scale:** n8n queue mode (Redis + workers) to lift the 1-vCPU ceiling.
- **Exactly-once processing:** an atomic idempotency claim for duplicates that arrive at the same moment.
- **Named owners:** assign each HubSpot task to a real person, not just a team.
- **A learning loop:** feed the Insights Agent's recommendations back into the Content Agent's prompt automatically.

Code, results and outputs github.com/Mayank-1024/Madison-Project

Built by Mayank Bhadrasen

Stack n8n + Claude Haiku 4.5